Grade 10 - Term 1: C6 & C7

Expected Monetary Value - Version 2

Learning Criterion

C6. Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff.

C7. Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

1 C6 & C7 Expected Monetary Value

When probabilities can be assigned to the possible states of nature, the Expected Monetary Value (EMV) criterion compares alternatives by combining each possible payoff with the probability of the state in which that payoff occurs.

Focus	Expected payoff
First step	Calculate the EMV of each alternative
Second step	Choose the highest EMV

Let $P(S_j)$ represent the probability that state of nature S_j occurs. All probabilities are nonnegative: $P(S_j) \geq 0$. They must satisfy

$$\sum_{j=1}^{n} P(S_j) = 1$$

In every exercise, choose an alternative before the external state is known. The same state probabilities apply to all alternatives. All table entries are monetary payoffs, and higher values are better. All payoffs and EMVs are in **thousand USD**.

For each alternative, calculate its Expected Monetary Value:

$$EMV(A_i) = \sum_{j=1}^{n} P(S_j) P_{ay}(A_i, S_j)$$

For two states of nature, this becomes

$$EMV(A_i) = P(S_1) P_{ay}(A_i, S_1) + P(S_2) P_{ay}(A_i, S_2)$$

Finally, choose the alternative with the highest expected monetary value:

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i)$$

Sensitivity Analysis

When the probabilities of the states are subjective, the preferred alternative may depend on the probability assigned to each state. Sensitivity analysis examines how the decision changes when these probabilities change. For two states of nature, let $p = P(S_1)$. Since the probabilities must sum to one,

$$P(S_1) = p, \qquad P(S_2) = 1 - p, \qquad 0 \leq p \leq 1.$$

The EMV of each alternative can then be written as a function of p:

$$EMV(A_i) = pP_{ay}(A_i, S_1) + (1-p)P_{ay}(A_i, S_2)$$

A possible switching threshold occurs when two alternatives have equal expected monetary value:

$$EMV(A_i) = EMV(A_k)$$

An equality is a decision threshold only if the tied alternatives attain the highest EMV. There may be no feasible threshold, or several thresholds. Decision thresholds divide the feasible probability range into regions in which different alternatives have the highest EMV. Therefore, the sensitivity analysis determines which alternative is preferred for each range of values of p and identifies any probability at which the decision-maker is indifferent between alternatives.

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2 Solved Example: Museum Gallery

A museum must choose an expanded gallery (A_1) or a compact gallery (A_2) before regional tourism is known. Tourism may be busy (S_1) or quiet (S_2). Visitor records give estimated probabilities of 70% and 30%, respectively.

Alternative	S_1	S_2
$P(S_j)$:	70%	30%
Expanded gallery (A_1)	62	8
Compact gallery (A_2)	39	26

All payoffs and EMVs are in thousand USD. Which gallery should the museum choose using Expected Monetary Value?

2.1 Solution

Step 0: Identify the probabilities The probabilities are $P(S_1) = 0.7$ and $P(S_2) = 0.3$. Both are nonnegative. Check their sum:

$$P(S_1) + P(S_2) = 0.7 + 0.3 = 1.$$

Step 1.1: Calculate the EMV of A_1

Multiply each payoff by its state's probability and add:

$$\begin{aligned} EMV(A_1) &= \sum_{j=1}^{2} P(S_j) P_{ay}(A_1, S_j) \\ &= (0.7)(62) + (0.3)(8) \\ &= 43.4 + 2.4 = 45.8. \end{aligned}$$

The expanded gallery has an EMV of 45.8 thousand USD, equivalent to \$45,800.

Step 1.2: Calculate the EMV of A_2

$$\begin{aligned} EMV(A_2) &= \sum_{j=1}^{2} P(S_j) P_{ay}(A_2, S_j) \\ &= (0.7)(39) + (0.3)(26) \\ &= 27.3 + 7.8 = 35.1. \end{aligned}$$

The compact gallery has an EMV of 35.1 thousand USD, equivalent to \$35,100.

Step 2: Apply the Bayes decision criterion

Compare both EMVs:

$$R^* = \max\{45.8, 35.1\} = 45.8,$$
$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i) = A_1.$$

Choose the expanded gallery: its expected monetary payoff is \$45,800, compared with \$35,100 for the compact gallery.

Extended payoff table

Alternative	S_1 (0.7)	S_2 (0.3)	EMV
A_1	62	8	$0.7(62) + 0.3(8) = 45.8$
A_2	39	26	$0.7(39) + 0.3(26) = 35.1$
Highest EMV			45.8

3 Guided Practice: Orchard Lease

A grower must lease a hillside orchard (A_1) or a valley orchard (A_2) before seasonal rainfall is known. Rainfall may be ample (S_1), moderate (S_2), or scarce (S_3). Weather records give estimated probabilities of 40%, 35%, and 25%, respectively. Payoffs and EMVs are in thousand USD.

	Rainfall		
Alternative	Ample (S_1)	Moderate (S_2)	Scarce (S_3)
Hillside (A_1)	96	34	-18
Valley (A_2)	68	48	24

All payoffs and EMVs are in thousand USD. Determine the decision using the Expected Monetary Value criterion.

Step 1: Calculate the EMV of each alternative

$$\begin{aligned} EMV(A_1) &= (0.4)(96) + (0.35)(34) + (0.25)(-18) \\ &= _____ \end{aligned}$$

$$\begin{aligned} EMV(A_2) &= (0.4)(68) + (0.35)(48) + (0.25)(24) \\ &= _____ \end{aligned}$$

Step 2: Apply the Bayes decision criterion

Choose the alternative with the highest expected monetary value:

$$EMV^* = \max\{_____, _____\} = _____$$

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i) = _____$$

4 Scaffolded Practice: Campsite Plan

A campsite owner must choose open pitches (A_1), covered pitches (A_2), cabins (A_3), or a fixed lease (A_4) before the summer weather is known. Weather records estimate dry weather (S_1) at 55% and wet weather (S_2) at 45%. The fixed lease guarantees the same payoff in either state. Payoffs and EMVs are in thousand USD.

	Weather	
Alternative	Dry (S_1)	Wet (S_2)
Open pitches (A_1)	94	-16
Covered pitches (A_2)	70	18
Cabins (A_3)	46	32
Fixed lease (A_4)	24	24

Step 1: Calculate the EMV of each alternative

Alt.	EMV
A_1	$(0.55)(94) + (0.45)(-16) =$ _____
A_2	$(0.55)(70) + (0.45)(18) =$ _____
A_3	$(0.55)(46) + (0.45)(32) =$ _____
A_4	$(0.55)(24) + (0.45)(24) =$ _____

Step 2: Apply the Bayes decision criterion

$$EMV^* = \max\{_____, _____, _____, _____\} = _____$$

$$A^*_{\text{Bayes}} = \arg\max_i EMV(A_i) = _____$$

5 Independent Practice: Translation Rights

A publisher must choose worldwide translation rights (A_1), selected language rights (A_2), regional rights (A_3), or a fixed license (A_4) before overseas interest in a novel is known. Interest may be strong (S_1), steady (S_2), or weak (S_3). Reader research estimates probabilities of 45%, 25%, and 30%, respectively. The fixed license guarantees the same payment in all states. Payoffs and EMVs are in thousand USD.

	Overseas interest		
Alternative	S_1	S_2	S_3
A_1	78	42	12
A_2	62	49	23
A_3	44	37	31
A_4	26	26	26

What alternative should the publisher choose using the Expected Monetary Value criterion?

6 Independent Practice: Irrigation Plan

A greenhouse cooperative must select intensive irrigation (A_1), balanced irrigation (A_2), or minimal irrigation (A_3) before the dry season. The external water supply may be abundant (S_1), adequate (S_2), limited (S_3), or restricted (S_4). Reservoir research estimates the probabilities as 25%, 30%, 25%, and 20%, respectively. Payoffs and EMVs are in thousand USD.

	Water supply			
Alternative	S_1	S_2	S_3	S_4
A_1	102	76	34	-14
A_2	79	63	41	21
A_3	52	47	39	29

What alternative should the cooperative choose using the Expected Monetary Value criterion?

7 Solved Example: Wildlife Hide

A reserve must choose an open hide (A_1) or a sheltered hide (A_2) before the season's visibility is known. Visibility may be clear (S_1) or misty (S_2). Let p be the reserve manager's subjective probability of clear visibility, based on judgment; mist has probability $1-p$, where $0 \le p \le 1$. Payoffs and EMVs are in thousand USD. Find how the preferred hide changes with p.

Alternative	S_1	S_2
$P(S_j)$:	p	$1-p$
Open hide (A_1)	68	8
Sheltered hide (A_2)	48	38

Step 1.1: Calculate the EMV of A_1

$$\begin{aligned} EMV(A_1) &= p(68) + (1-p)(8) \\ &= 68p + 8 - 8p = 60p + 8. \end{aligned}$$

Its expected payoff is $60p + 8$ thousand USD.

Step 1.2: Calculate the EMV of A_2

$$\begin{aligned} EMV(A_2) &= p(48) + (1-p)(38) \\ &= 48p + 38 - 38p = 10p + 38. \end{aligned}$$

Its expected payoff is $10p + 38$ thousand USD.

Step 2: Find the threshold probability

Set $EMV(A_1) = EMV(A_2)$:

$$\begin{aligned} 60p + 8 &= 10p + 38 \\ 50p &= 30 \\ p &= \frac{30}{50} = \frac{3}{5} = 0.6. \end{aligned}$$

Step 3: Perform the sensitivity analysis

The difference is $EMV(A_1) - EMV(A_2) = 50p - 30$. Compare on both sides of $p = \frac{3}{5}$:

$$p < \tfrac{3}{5} \;\Rightarrow\; EMV(A_1) < EMV(A_2) \;\Rightarrow\; A_2,$$

$$p > \tfrac{3}{5} \;\Rightarrow\; EMV(A_1) > EMV(A_2) \;\Rightarrow\; A_1,$$

$$p = \tfrac{3}{5} \;\Rightarrow\; EMV(A_1) = EMV(A_2) \;\Rightarrow\; \text{tie}.$$

At the threshold both EMVs are 44 thousand USD. Thus:

$$A^*_{\text{Bayes}} = \begin{cases} A_2 & 0 \le p < \frac{3}{5}, \\ \text{Indifferent} & p = \frac{3}{5}, \\ A_1 & \frac{3}{5} < p \le 1. \end{cases}$$

Choose the sheltered hide below a 60% subjective chance of clear visibility, and the open hide above it; either is optimal at 60%.

8 Guided Practice: Art Workshop

An art school must book a courtyard (A_1) or an indoor studio (A_2) before workshop-day weather is known. Weather may be dry (S_1) or rainy (S_2). Let p be the coordinator's subjective probability of dry weather, based on judgment; rain has probability $1-p$, with $0 \le p \le 1$. Payoffs and EMVs are in thousand USD. Find the preferred venue for every p and the probability at which the coordinator is indifferent.

Alternative	S_1	S_2
$P(S_j)$:	p	$1-p$
Courtyard (A_1)	58	6
Indoor studio (A_2)	43	31

Step 1.1: Calculate the EMV of A_1

$$EMV(A_1) = p(58) + (1-p)(6)$$
$$= _____$$
$$= _____$$

Courtyard EMV: _____ thousand USD.

Step 1.2: Calculate the EMV of A_2

$$EMV(A_2) = p(43) + (1-p)(31)$$
$$= _____$$
$$= _____$$

Indoor studio EMV: _____ thousand USD.

Step 2: Find the threshold probability

Set the two EMVs equal:

$$EMV(A_1) = EMV(A_2)$$

$$_____ = _____$$

$$_____\, p = _____ \quad \longrightarrow \quad p = _____$$

Indifference occurs at the dry-weather probability _____.

Step 3: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

$p <$ _____ $\Rightarrow$ $EMV(A_1) < EMV(A_2)$ $\Rightarrow$ choose _____,

$p >$ _____ $\Rightarrow$ $EMV(A_1) > EMV(A_2)$ $\Rightarrow$ choose _____,

$p =$ _____ $\Rightarrow$ $EMV(A_1) = EMV(A_2)$ $\Rightarrow$ _____.

$$A^*_{\text{Bayes}} = \begin{cases} _____, & 0 \le p < _____, \\ _____, & p = _____, \\ _____, & _____ < p \le 1. \end{cases}$$

9 Independent Practice: Apiary

A beekeeper must choose a meadow apiary (A_1) or woodland apiary (A_2) before flowering conditions are known: favorable (S_1) or poor (S_2). Let p be the beekeeper's subjective probability of favorable conditions, based on judgment; poor conditions have probability $1-p$, $0 \le p \le 1$. Payoffs and EMVs are in thousand USD. Find the preferred apiary for every p, including indifference.

Alternative	S_1	S_2
$P(S_j)$:	p	$1-p$
Meadow apiary (A_1)	76	12
Woodland apiary (A_2)	51	32

10 Dominance Practice: Archive

An archive must choose full service (A_1) or basic service (A_2) before grant funding is known: ample (S_1) or limited (S_2). Let p be the director's subjective probability of ample funding, based on judgment; limited funding has probability $1-p$, $0 \le p \le 1$. Payoffs and EMVs are in thousand USD. Which service is preferred for every p? Does a feasible switching threshold exist? If none exists, explain why.

Alternative	S_1	S_2
$P(S_j)$:	p	$1-p$
Full service (A_1)	57	44
Basic service (A_2)	49	36

11 Practice: Vineyard Layout

A vineyard must choose hillside vines (A_1), mixed plots (A_2), or sheltered vines (A_3) before growing conditions are known. Conditions may be favorable (S_1) or adverse (S_2). Let p be the grower's subjective probability of favorable conditions, based on judgment; adverse conditions have probability $1 - p$, where $0 \le p \le 1$. Payoffs and EMVs are in thousand USD. Compare all three EMV functions and give the preferred layout over the whole feasible interval, including all optimal ties. Check whether each pairwise intersection actually changes the optimal decision.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Hillside vines (A_1)	86	6
Mixed plots (A_2)	66	26
Sheltered vines (A_3)	46	36

12 Practice: Archaeological Survey

A survey firm must choose extensive fieldwork (A_1) or a limited survey (A_2) before land-access permits are decided. Access may be full (S_1), partial (S_2), or denied (S_3). Let p be the director's subjective probability of full access, based on judgment, with $0 \le p \le 1$. Conditional on access not being full, the director assigns probability $\alpha = 0.25$ to partial access and $1 - \alpha = 0.75$ to denial. Thus the state probabilities are p, $0.25(1 - p)$, and $0.75(1 - p)$. Payoffs and EMVs are in thousand USD. Determine the preferred survey for every p and identify the indifference threshold.

Alternative	S_1	S_2	S_3
$P(S_j)$:	p	$0.25(1 - p)$	$0.75(1 - p)$
Extensive (A_1)	92	60	12
Limited (A_2)	62	44	28

13 Practice: Mineral Exploration

An exploration firm must choose deep drilling (A_1), mixed drilling (A_2), shallow drilling (A_3), or a surface survey (A_4) before the site's mineral grade is known. Grade may be rich (S_1) or lean (S_2). Let p be the geologist's subjective probability of rich grade, based on judgment; lean grade has probability $1 - p$, with $0 \le p \le 1$. Payoffs and EMVs are in thousand USD. Compare all four EMV functions and give the preferred plan over the whole feasible interval, including all optimal ties. Check whether each pairwise intersection actually changes the optimal decision.

Alternative	S_1	S_2
$P(S_j)$:	p	$1-p$
Deep drilling (A_1)	78	8
Mixed drilling (A_2)	73	23
Shallow drilling (A_3)	63	33
Surface survey (A_4)	48	38

14 Practice: Marina Berth Agreement

A marina must choose a seasonal berth agreement (A_1) or an annual berth agreement (A_2) before coastal sailing conditions are known. Conditions may be excellent (S_1), fair (S_2), or stormy (S_3). Let p and q be the manager's subjective probabilities of excellent and fair conditions, based on judgment. Stormy conditions have probability $1 - p - q$. The feasible domain is

$$p \ge 0, \qquad q \ge 0, \qquad p + q \le 1.$$

Payoffs and EMVs are in thousand USD. Construct both EMV functions, find the indifference line, and determine the region in the feasible domain where each agreement is preferred, including the tie case.

Alternative	S_1	S_2	S_3
$P(S_j)$:	p	q	$1-p-q$
Seasonal berths (A_1)	84	56	9
Annual berths (A_2)	54	41	29